

T1 Python Project: News Headline Tracker

Learn how to store, search, and filter headlines using Python Lists, Loops, and Conditionals!

Today's Mission: Become a News Detective!

Imagine you are a news detective! Your mission is to write a Python program that scans daily news headlines and automatically detects exciting topics like **ISRO**, **Cricket**, or **Weather** in seconds.

1. How Our News Tracker Works

To build a news tracker, we combine three fundamental concepts in programming:

Concept	Real-World Analogy	Python Role
Python List	A big newsbag holding newspaper clipping	Stores multiple daily news headlines as text items in one variable.
For Loop	Flipping through TV news channels one by one	Iterates through the list, inspecting each headline sequentially.
If Statement	A detective checking for specific clues	Evaluates if a keyword (e.g., "weather") exists in the headline.
String Case Handling	Ignoring uppercase/lowercase in handwriting	Uses <code>lower()</code> so searches match regardless of letter casing.

2. Starter Code: Storing News Headlines

Python lists are enclosed in square brackets `[]`, with text items enclosed in quotes `""` and separated by commas.

```
# Starter List of Headlines
headlines = [
    "ISRO launches new rocket",
    "IPL cricket updates today",
    "Delhi weather forecast",
    "Neeraj Chopra wins gold",
    "Dussehra celebrations begin"
]
```

3. Searching Headlines with User Input

We allow the user to type a search word using `input()`, and then check each headline in the list:

```
# Get user search keyword
search_word = input("Enter keyword to search (e.g., Weather, ISRO): ")
```

```
for headline in headlines:
    # Check if search keyword exists in headline (ignoring case)
    if search_word.lower() in headline.lower():
        print(f"ALERT! Found match: '{headline}'")
```

4. Bug-Buster Challenge: Common Pitfalls

Watch out for these frequent coding errors when building your scanner:

- **Missing Colons:** Python requires a colon `:` at the end of `for` loops and `if` statements.
- **Case Sensitivity:** Searching for `"isro"` will NOT match `"ISRO"` unless you convert both strings to lowercase using `.lower()`.
- **Indentation:** The code blocks inside loops and `if` statements must be indented (4 spaces).

Summary & Key Takeaways

- Lists organize data for easy scanning and retrieval.
- Loops automate repetitive scanning tasks without duplicate code.
- Dynamic user inputs (`input()`) make programs interactive and responsive.

Practice Questions for Students

Complete the following exercises to test your Python knowledge. (No answers provided)

Question 1: Multiple Choice

Which Python data structure is used to store multiple headlines together in a single variable?

A) Integer B) List C) Boolean D) Float

Question 2: Syntax Identification

Identify the syntax error in the following code snippet and explain what is missing:

```
for item in headlines
    if "cricket" in item.lower()
        print(item)
```

Question 3: String Manipulation

Why do we use the `.lower()` method on both the search keyword and the headline text during a search operation?

Question 4: Code Completion

Complete the missing parts (indicated by `___`) to prompt the user for input and count how many headlines contain that keyword:

```
user_query = ___("Enter topic: ")
match_count = 0

for headline in headlines:
    if user_query.lower() ___ headline.lower():
        match_count = match_count + 1

print("Total matches found:", match_count)
```

Question 5: Mini Coding Task

Write a Python loop that iterates through the `headlines` list and prints only the headlines that have more than 20 characters in length (Hint: use `len()`).